Applied geochemical and reactive transport modelling using the HPx framework – Tutorial for modelling cement systems

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GLOSSARY

Case	When a simulation contains an input table and parameter values are defined in that table, a case represents one particular calculation in the simulation with that set of parameters. Calculation results can be saved in a separate subfolder in the simulation folder. *See* Tutorial 4-3 Define cases in a simulation
Parametric study	Based on a global input table, a series of calculations will be performed in which selected numerical values will vary automatically based on (i) start or end position of a parameter vector, (ii) parameter vectors listed in a table, (iii) centered around a parameter vector, or (iv) gridded parameter vectors. *See* Tutorial 4-5 Blending OPC and GBFS along a reaction path

PREFACE

1. Introduction

1.1 Cement chemistry

2. NOTES ON CHEMISTRY OF CEMENT SYSTEMS

2.1 Cement systems

In this tutorial, the term cement systems refers to the oxides that are defined to form cement hydrates and secondary phases. For example, a Ca-Si-C system uses only the oxides of Ca, Si and C and related hydrates such as portlandite, C-S-H and calcite. When hydration is simulated, also only those oxides are released during clinker hydration.

Shows the systems implemented in the module.

Nr	
Basic systems (1-30)	
7	Ca Si Al S Mg C
9	Ca Si Al S Mg C Cl
11	Ca Si Al Fe S Mg C
13	Ca Si Al S Mg Na K
17	Ca Si Al S Mg Na K C Cl
19	Ca Si Al Fe S Mg Na K C Cl
Systems with N (31-60)	
43	Ca Si Al S Mg Na K N
47	Ca Si Al Fe S Mg Na K C Cl N

3. DESCRIPTION OF THE CEMENT MODULE

3.1 Download and install

The cement module is available on github via this link. One need to download a zip file, unzip it and bring the module .

A module should be stored in the general Module directory of HPx (HYDRUS 5.x) and can be used in several projects. Tutorial 3-1 shows how to obtain the module and place it in the correct folder such that it can be activated in your projects.

TUTORIAL 3-1 INSTALL CEMENT MODULE

Description

To add the cement module to HPGeochemistry

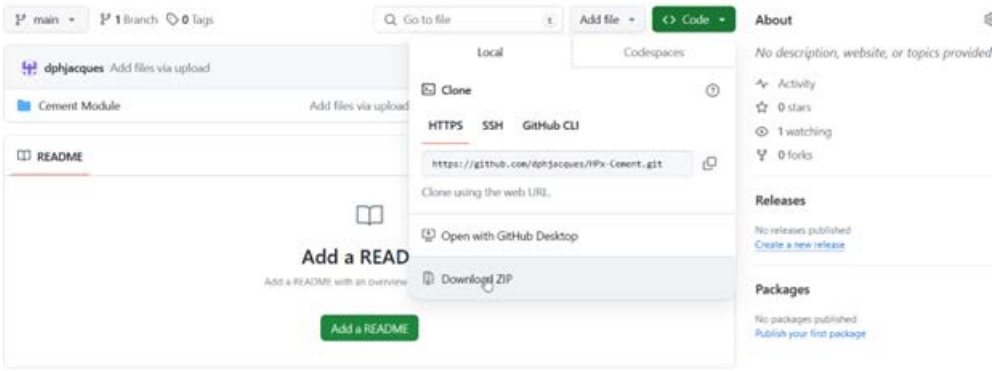
You will learn to

Add a module to HPGeochemistry

Step 1 – Download Cement Module

Go to the github repository via **link**

Download the repository via Code and Download ZIP



Step 2 – Find the general folder for Modules

The folders and files under Cement Module should be placed in the default Module folder of HPx/HYDRUS 5.x. It will then be available to be used it in different projects.

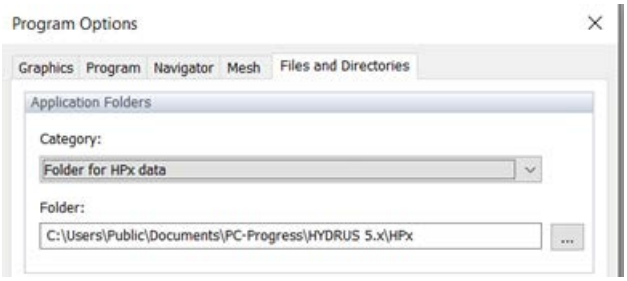
The default folder can be found via

(i) HYDRUS 5.0 GUI

Go to `Options->Program Options`...

Go to `Files and Directories` tab

Change the `Category` of the `Application Folders` to `Folder for HPx data`



The default folder is a subfolder of this folder. In this case, it is:

`C:\Users\Public\Documents\PC-Progress\HYDRUS 5.x\HPx\Modules`

(ii) HPGeochemistry GUI

Go to `Settings and Preferences`

Go to the `Settings` tab

Go to the `Databases` subtab

The folder in the textbox "`Default folder for database groups`" indicates the default folder for databases. The parent folder gives also the path to subfolder `Modules`. Thus, in case you see



the default folder for modules is

`C:\Users\Public\Documents\PC-Progress\HYDRUS 5.x\HPx\Modules`

Step 3 – Place the cement module in the general folder

Make a subfolder `Cement Module` in the default folder for modules

Unzip the downloaded

Place the subfolder in the folder `HPx-Cement-main\Cement Module` in the subfolder `Cement Module`

END TUTORIAL 3-1

3.2 Set-up HPGeochemistry

The tutorial for setting up a project, add databases and modules script is limited to a list of instructions. The reader is referred to XX for more details on the basic use of HPGeochemistry.

All building blocks are present in the downloaded database files and cement module.

TUTORIAL 3-2 CREATE MANAGED PROJECT WITH THE CEMENT MODULE

Description

Create the project for this tutorial and add the cement module in the project

You will learn to

To create a managed project

To add a module to the project

Depends on

Tutorial 3-1 Install CEMENT module

Step 1 – Create project

Click `Project Manager` button 

Click `Create new project` button 

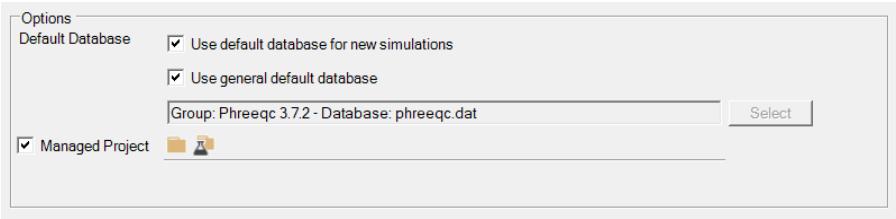
The `Metadata` dialogue window appears.

Give a name to your project

Step 2 – Add cement module to the project

A prerequisite to use a module in HPGeochemistry is that it is a managed project:

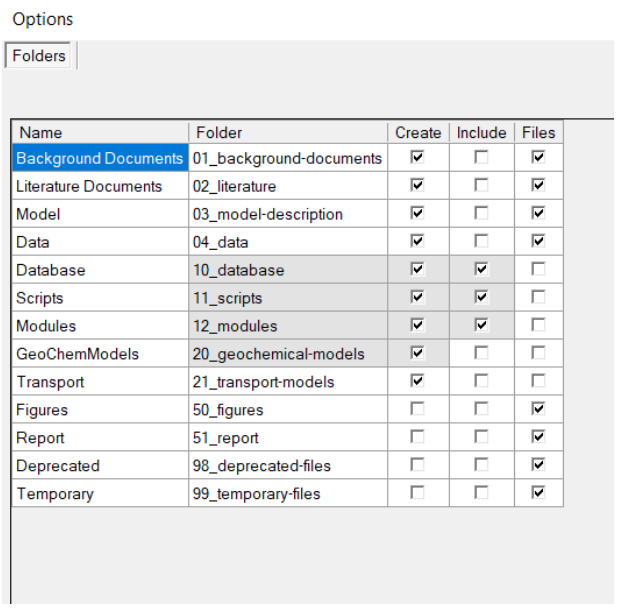
Ensure that the Managed Project is checked



For the Cement Module, folders for data, database and modules are required.

Open the `Folder Manager` for a managed project 

Ensure that the folder `Data` (`04_data`) is checked

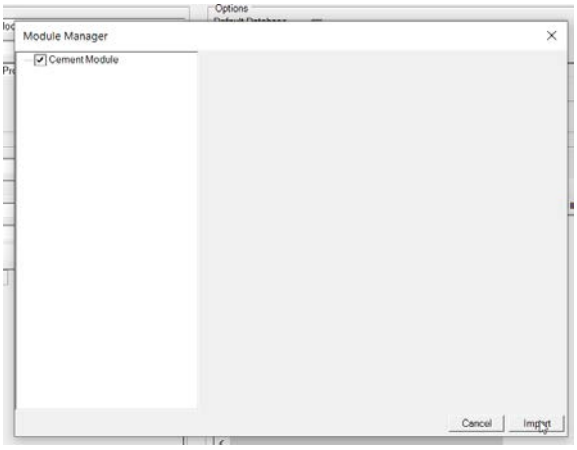


OK

Open the `Module Manager` for a managed project 

Check `Cement Module`

Click `Import`



Click `OK`

END TUTORIAL 3-2

4. THERMODYNAMIC MODELLING OF CEMENT HYDRATES

4.1 Modelling CEM-I hydrate assemblage and pore water composition

4.1.1 Calculation of the hydrate assemblage for a CEM-I oxide composition

TUTORIAL 4-1 CALCULATE THE CEMENT HYDRATES ASSEMBLAGE AND PORE WATER COMPOSITION FOR A CEM I

Description

Calculate the hydrates assemblage and pore water composition of the hydration of 100 g CEM-I with a water/cement ratio of 0.5 and at a temperature of 25°C. The CEM I oxide composition is taken from the Lothenbach and Winnefeld (2006) (their table 1). Calculations are done in the Ca-Si-Al/Fe-S-Mg-C-Na-K system.

You will learn to

Start a simulation from a template in a module

Depends on

Tutorial 3-2 Create managed project with the cement module

Step 1 – Create simulation from template for the calculation of the hydrates assemblages

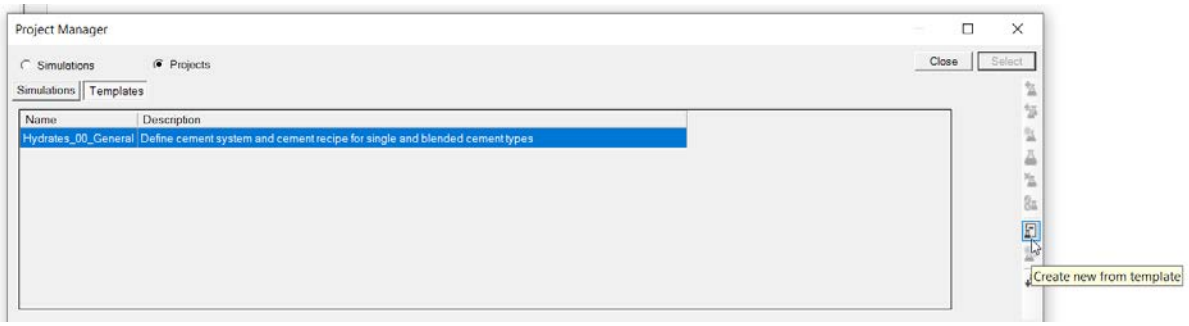
Go to `Project Manager`

Select the project in the bottom panel

Go to `Templates` tab in the top panel

Select the template "`Hydrates_00_general`"

Click the "`Create new from template`" button



The `Metadata` dialogue window opens.

Define the name of the simulation (`CEM I`)

Click `OK`

The Project Manager is active again.

Select the simulation (CEM I) and click `Select`

Step 2 – Define the problem

Only information in the input tables needs to be changed if values are different from the default values. All instructions and data are included in the simulation via the template.

Go to the `Global Definitions` input tab

Go to the `Global Variables`

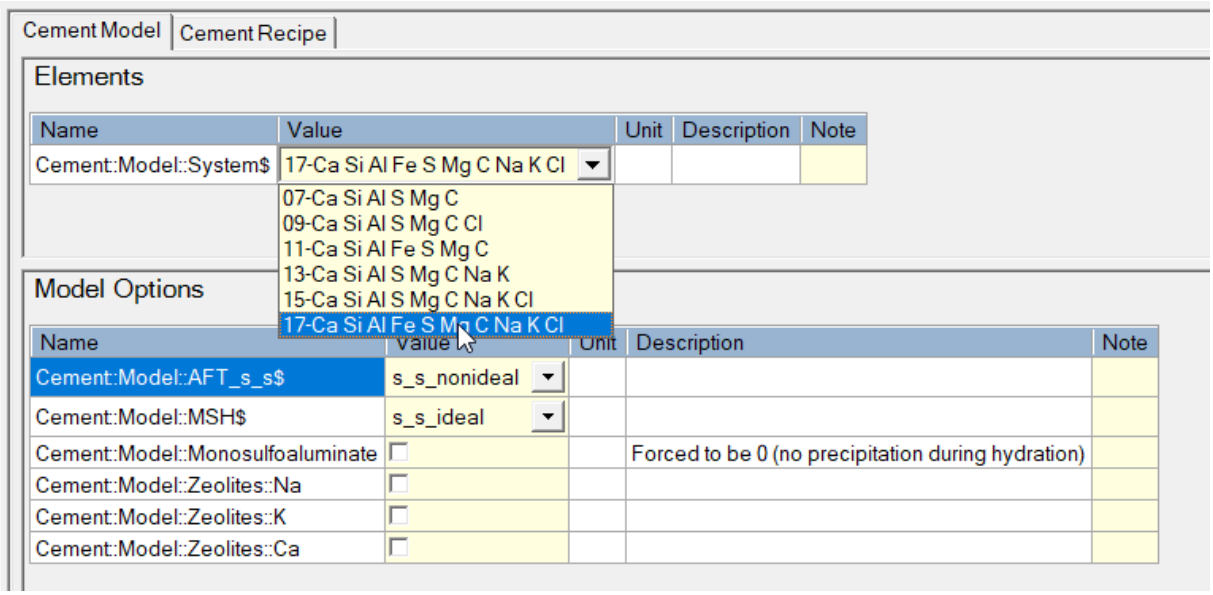
The template consists of two input tables: "`Cement Model`" and "`Cement Recipe`" to, respectively, select the cement model and define the cement recipe.

Go to the `Cement Model` tab

The first group "Elements" defines which elements are to be considered in the calculations. Selection is done via the drop down menu for the variable `Cement::Model::System$`.

Select system 17 from the drop down menu. Note that this system also contains Cl but is neglected in this tutorial (no Cl in the clinker composition)

The second group "`Model Options`" allows to select some specific minerals to be included or excluded from the models – default selections are used in this tutorial.



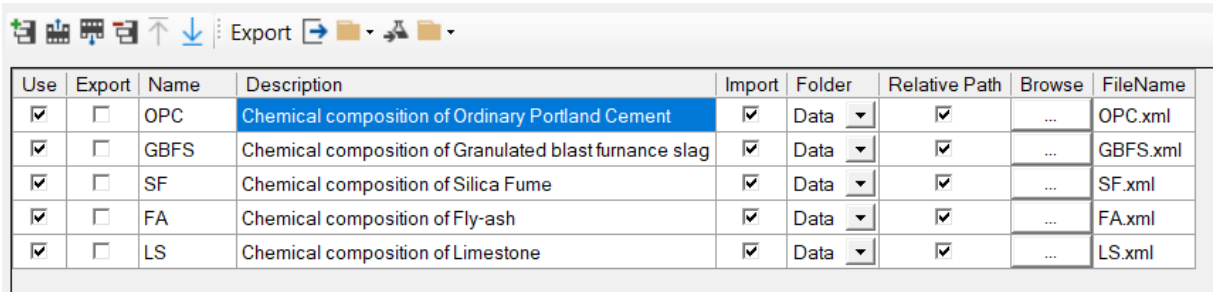
Go to the `Cement Recipe` tab

In the first group "Properties", the temperature (°C), water/cement ratio (-) and amount of cement (g) are given. In this tutorial, the default values are used.

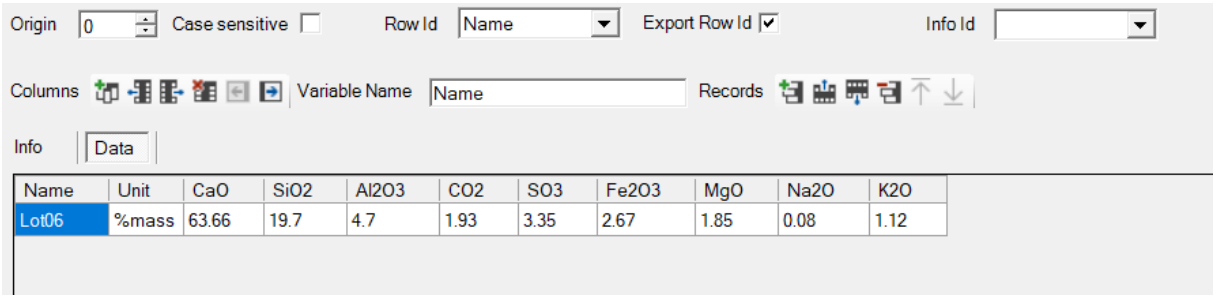
The five other groups represent the ordinary Portland cement (OPC) clinker and following supplementary cementitious materials (SCMs): granulated blast furnace slag (GBFS), silica fume (SF), fly-ash (FA) and limestone (LS). For each material, it is possible to define oxide composition, the amount (mass%), the reacted amount (% reacted) and the bulk density (kg/dm^3).

The composition of the OPC clinker and the SCMs are given via data tables defined in the DataBlock. Each record in the data table forms an item in the drop down list. Beside the default options, additional compositions can be defined (see Tutorial 4-2).

Go to `DataBlock`



Select `OPC` data block in the top panel. Information in that data block is shown in the bottom panel:



For each of the cement constituents (OPC and SCMs), a data table is available. In each of these data table, an oxide composition is given by default (see example for OPC). The indicator in the column `Name` is used in the drop down list. The reference to the oxide composition is listed in the `Info` tab. Other compositions can be added (see Tutorial 4-2).

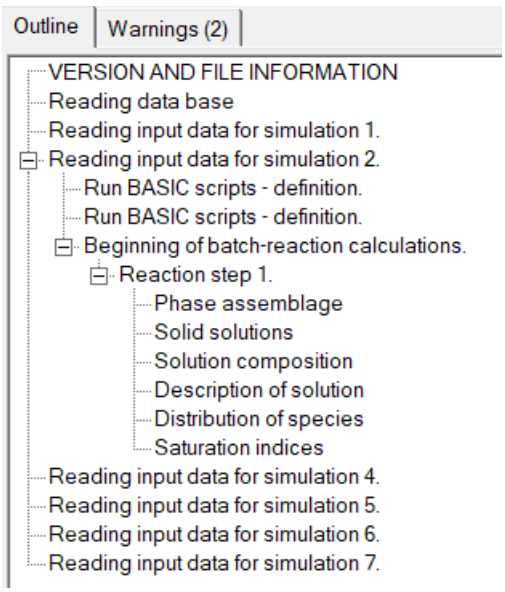
Run the problem 

Step 3 – Check the output

Go to `phreeqc.out` file

Go to `Outline` tab

Go to `Reading input data for simulation 2` -> `Beginning of batch-reaction calculation`



Note first that there are warnings. This is because Cl-containing hydrates are in the hydrate assemblage (due to the selection of system 17), but there is no Cl in the input.

The output file gives the standard phreeqc output related to a batch reaction calculation. All geochemical information can be obtained from this file. The template also produces a special output file with the most important information for the cement system.

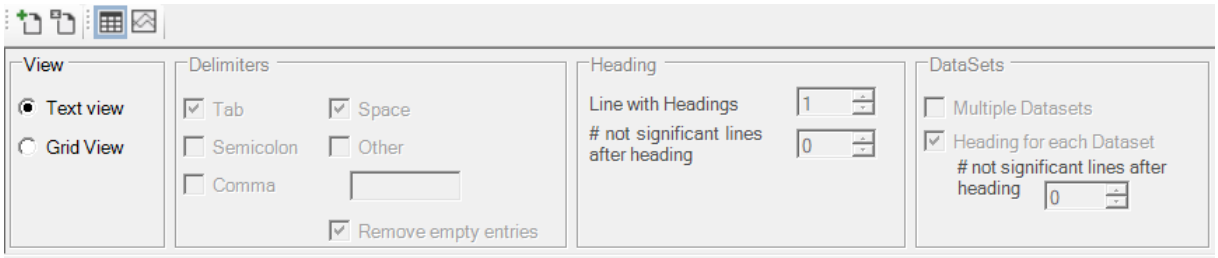
Go to the Data File output workspace

Click Open Data File

Open the file "Cement_Hydration_1001.out"

Go to the View options

Select Text View



Cement_Hydration_1001.out

```
CEMENT RECIPE
                        OPC clinkers            100 weight%
                         Reacted OPC            100 % reacted

                    Amount of cement            100
                           w/c ratio           0.50
                         Temperature             25

CONTENT
              Name          Moles
               CaO       1.14e+00
              SiO2       3.28e-01
             Al2O3       4.61e-02
               SO3       4.18e-02
               CO2       4.39e-02
               MgO       4.59e-02
             Fe2O3       1.67e-02
               Na2O      1.29e-03
               K2O       1.19e-02

HYDRATED PHASES

              Name          Moles          Grams         Volume

      Portlandite       4.17e-01          30.90          13.79
              Cal       3.49e-02           3.50           1.29
     hydrotalcite       1.15e-02           5.09           2.53
    monocarbonate       5.04e-03           2.87           1.32
              AFT       4.56e-02          18.95          10.68
              CSH       4.28e-01          61.52          27.22
    HydrogarnetFe       3.11e-02          13.35           4.53
              MSH       9.51e-16           0.00           0.00
```

In this view, you find information on the cement recipe, the oxide composition, the hydrated phases and the pore water composition, calculated volumes, and information on solid solution composition.

END TUTORIAL 4-1

TUTORIAL 4-2 CHANGE THE CLINKER OXIDE COMPOSITION OF OPC

Description

Calculate the hydrates assemblage and pore water composition of the hydration of 100 g CEM-I with a water/cement ratio of 0.5 and at a temperature of 25°C with the oxide composition as defined in Table 4 in Glosser et al. (2021).

	SiO_2	Al_2O_3	CaO	Fe_2O_3	MgO	SO_3	Na_2O	K_2O
	20.90	3.62	61.63	3.20	3.20	2.13	0.17	0.53

You will learn to

Add a new clinker oxide composition of OPC to a datablock

Depends on

Tutorial 4-1 Calculate the cement hydrates assemblage and pore water composition for a CEM I.

Step 1 – Add the oxide clinker composition to the datablock

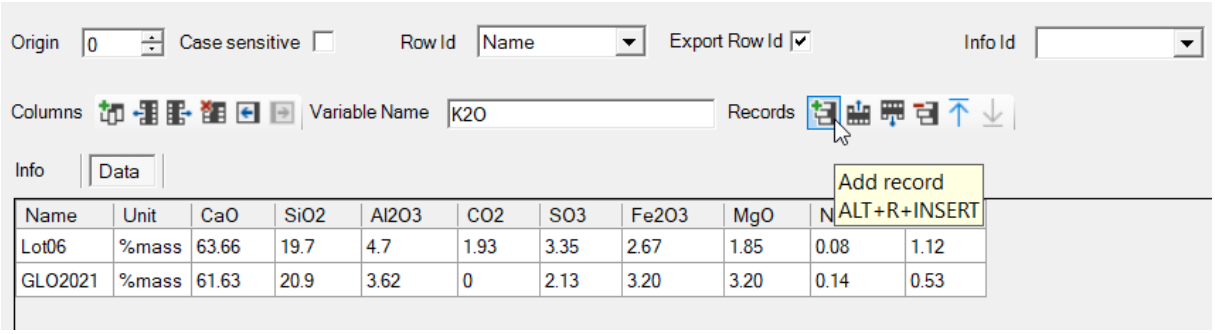
Open the simulation that was created in Tutorial 4-1 via the Project Manager or via the drop down list of previously opened simulations.

Go to `Global Definitions`

Go to `Datablock`

Select the datablock `OPC`

The selected datablock appears in the bottom panel – new records/row can be added.



Add a new record at the bottom of the table

Give a name (e.g., `GLO2021`)

Add the oxide composition

Step 2 – Select, Run and Output

Go to `Global Variables`

Go to `Cement Recipes`

Select `GLO2021` in dropdown menu `Cement::Clinker::Reference$`

Run the problem

Check output.

END TUTORIAL 4-2

4.1.2 Reaction path model with varying w/c ratio

4.1.3 Reaction path model with varying temperature

4.2 Modelling CEM-II and CEM-III hydrated phases and pore water composition

4.2.1 Replacement OPC with SCM

TUTORIAL 4-3 CALCULATE THE CEMENT HYDRATES ASSEMBLAGES WHEN PART OF OPC IS REPLACED BY GBFS

Description

Calculate the hydrates assemblage and pore water composition of the hydration of 100 g CEM-I with a water/cement ratio of 0.5 and at a temperature of 25°C where 10% of OPC is replaced by GFBS. The CEM I oxide composition is taken from the Lothenbach and Winnefeld (2006) (their table 1); the GBFS oxide composition from Adu-Amankwah et al. (2017). Calculations are done in the Ca-Si-Al/Fe-S-Mg-C-Na-K system.

You will learn to

Define different cases in one simulation

Depends on

Tutorial 4-1 Calculate the cement hydrates assemblage and pore water composition for a CEM I

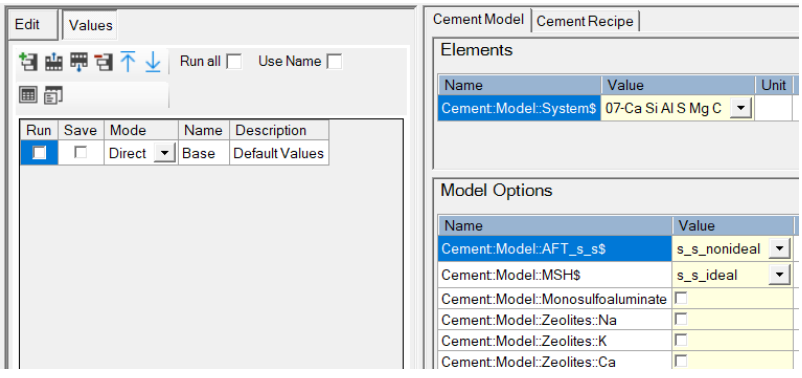
Step 1 – Create the case(s) in the simulation

Open the simulation that was created in Tutorial 4-1 via the Project Manager or via the drop down list of previously opened simulations.

Go to the `Global Variable` Tab

Go to `Global Variables`

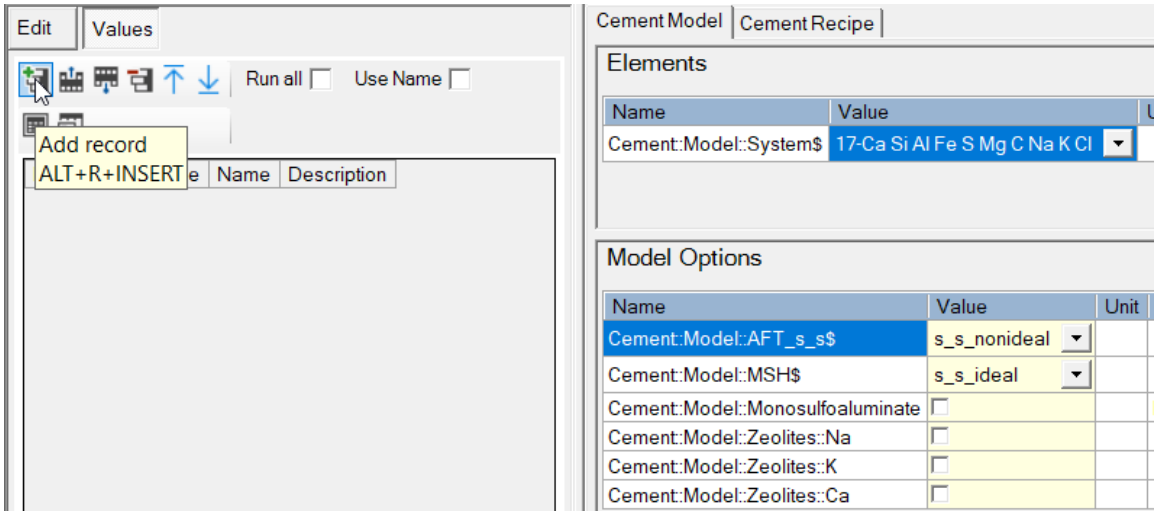
Cases are listed in the left panel



In a re-opened simulation, the table contains one case called Base by default. In a newly made simulation, this table is still empty, and cases can be added by the procedure described below.

Ensure that `Use Name` is checked. The subfolders containing the cases will have the name printed in the column `Name`.

Click to add a case – and add a case respectively before or after the selected case.



Ensure that you have two cases

Call the first case "CEMI" and check the box in the column with heading Save

Call the first case "CEMII/A-S" and check the box in the column with heading Save.

The first case uses the default values in the table, i.e. 100% OPC.

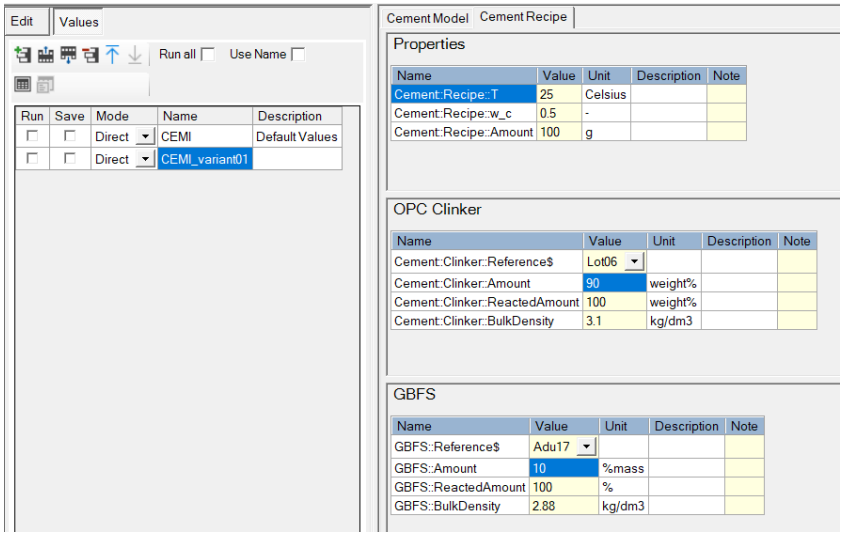
Select the second case (just select any cell in the second row)

Go to the `Cement Recipe` tab

Change following variables:

`Cement::Clinker::Amount` to 92

`GBFS::Amount` to 8



can be added via the left panel of the `Global Variables` Table

END TUTORIAL 4-3

The reactivity of a SCM is defined as the maximum amount of the SCM that is capable in participating in the chemical reactions – crystalline components generally do not react in cementitious systems (Glosser et al., 2021). Alternatively to assuming full reactivity of the SCM (e.g. in Tutorial 4-3), it can be assumed that only part of the SCM is reactive. Following the approaches proposed by Glosser et al. (2021) for fly-ash, the alternatives are for the case where the SCM composition is defined in terms of bulk content of oxides:

- An overall reactivity is defined; each oxide content is multiplied with the reactivity.
- An oxide-specific reactivity is defined; the oxide content is multiplied with the oxide specific reactivity.

Methods for obtaining the overall of the oxide-specific reactivity of fly-ash is given in Glosser et al. (2021).

For each SCM in the Cement Recipe table, the type of reactivity of the is specified via the drop down menu *SCM::Reactivity* (*SCM* is e.g. GBFS, FA, …) (Figure 4-1):

- **Option "Full Reactivity"** – assuming 100% reactivity of all oxides.
- **Option "Bulk Reactivity"** – assuming the same reactivity of all oxides. The reactivity is defined in data blocks with the oxide composition of the SCM in the column `Rtot` – see section 8.2.2.1.
- **Option "Oxide Specific"** – assuming an oxide-specific reactivity. The reactivity is defined in data blocks with the oxide composition of the SCM in the columns R*`oxide`* (*`oxide`* = CaO, SiO_2, …) – see section 8.2.2.1.

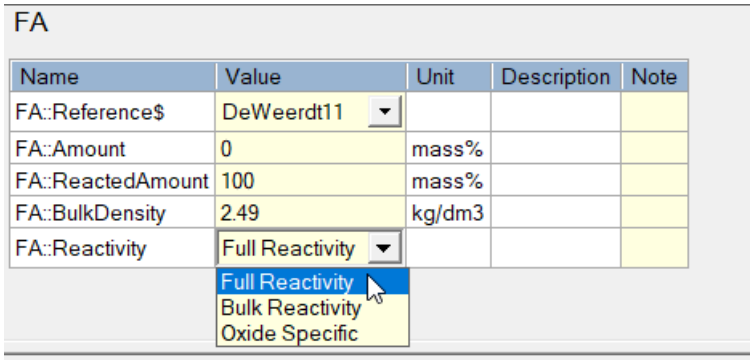


Figure 4-1 – Defining the reactivity of a SCM – Example for FA.

Note that the variable *SCM*`::ReactedAmount` parameter …expressed in %.

TUTORIAL 4-4 BLENDING OPC AND FA WITH DIFFERENT REACTIVITY OPTIONS FOR OXIDES IN FA

Description

Compare the hydrate assemblages of a blended cement with 90% OPC and 10% fly-ash (100 g cement, 0.5 water to cement ratio, 25 °C) assuming full reactivity, bulk reactive and oxide-specific reactive for fly-ash as given by Glosser et al. (2021) for fly-ash 4 in their Table 4 and their oxide composition as given in their Table 4 (see Tutorial 4-2).

The table below show the different relevant values for fly-ahs 4:

B – the oxide content (in weight%)

*DOR** - the weight percentage of reactive oxides calculated based on their method to define the bulk reactivity; the bulk reactive *Rtot* is also given in the table (20%°

$DOR_{\wedge H}$* - the weight percentage oxide of reactive oxides calculated based on their method to define the oxide specific reactivity; the oxide-specific reactive is given in the line *Roxide*

	SiO_2	Al_2O_3	CaO	Fe_2O_3	MgO	SO_3	Na_2O	K_2O
B	39.62	16.58	20.92	6.15	4.27	4.09	2.37	0.45
DOR*	7.92	3.32	4.18	1.23	0.85	0.82	0.47	0.09
DoR_{pH}*	6.02	15.04	9.07	5.49	0	3.62	2.21	0.45
Rtot	20	20	20	20	20	20	20	20
Roxide	15	91	43	89	0	89	93	100

You will learn to

Define the different types of reactivity of SCM

Depends on

Tutorial 4-3 Calculate the cement hydrates assemblages when part of OPC is replaced by GBFS

Step 1 – Add the information of the composition and reactivity of the fly-ash

Open the simulation that was created in Tutorial 4-1 via the Project Manager or via the drop down list of previously opened simulations.

Go to the `Global Variable` Tab

Go to Datablock

Select the datablock FA – The datablock for a SCL contains columns for the oxide weights, the bulk reactivity (Rtot) and the specific oxide reactivity (R*oxide*). Rtot and Roxide are in % of the bulk oxide content.

Add the information for GLO2021 with the information from the table above.

Step 2 – Add a case, adapt the cement recipe and select the SCM reactivity model

Go to Global Varaibles

Add a new case (e.g., CEM-II/A-V)

Change the OPC and FA content

Select in the drop down menu the reactivity model for FA (see Figure 4-1)

Run, check the output

Repeat for other reactivity models

4.2.2 Reaction path model with replacements

TUTORIAL 4-5 BLENDING OPC AND GBFS ALONG A REACTION PATH

Description

Calculate the hydrates assemblage and pore water composition of the hydration of 100 g CEM-I with a water/cement ratio of 0.5 and at a temperature of 25°C when OPC is replace by GBFS up to 60%. The CEM I oxide composition is taken from the Lothenbach and Winnefeld (2006) (their table 1); the GBFS oxide composition from Adu-Amankwah et al. (2017). Calculations are done in the Ca-Si-Al/Fe-S-Mg-C-Na-K system.

You will learn to

Include a parametric study

Depends on

Tutorial 3-2 Create managed project with the cement module

Step 1 – Create the simulation from the template to blend materials

Go to `Project Manager`

Select the project in the bottom panel

Go to `Templates` tab in the top panel

Select the template "`Hydrates_10_Bleding_general`"

Click the "`Create new from template`" button

The `Metadata` dialogue window opens.

Define the name of the simulation (`Blending`)

Click `OK`

The Project Manager is active again.

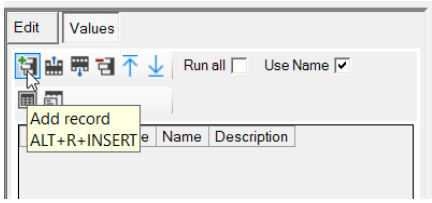
Select the simulation (Blending) and click `Select`

Step 2 – Create the reaction path of blending

Go to `Global Definitions`

Go to `Global Variables`

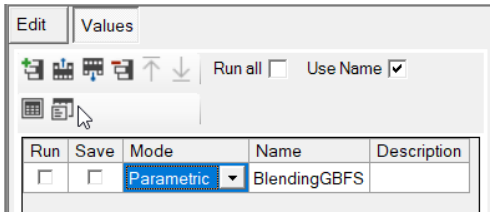
Add a case



Give the case a name (`BlendingGBFS`)

Select the `Parametric` Mode

Click on `Select Mode` ( ) icon

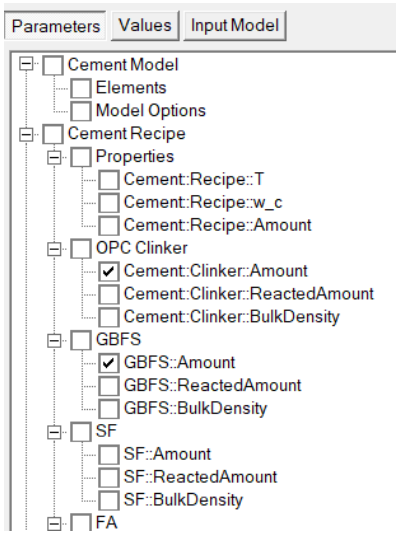


At the right side, the panel to define a parametric study is visible. There are three panels.

The first panel serves to select the variables included in the parametric study. It consists of all numerical variables in the global input table; Only selection at the third level are relevant.

Select `Cement::Clinker::Amount`

Select `GBFS::Amount`



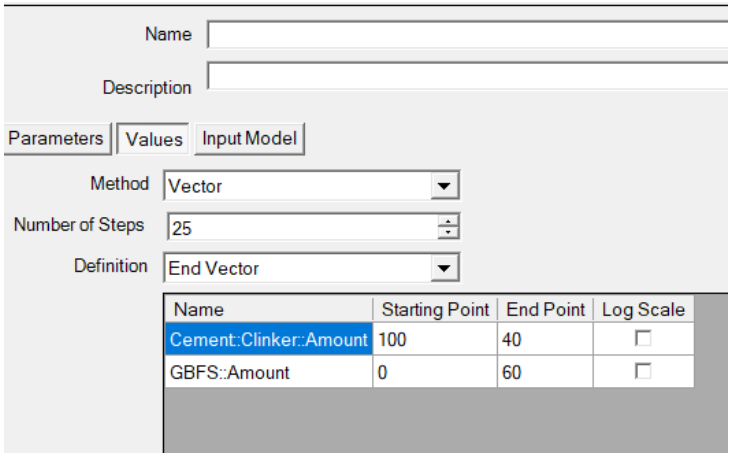
Go to the second panel (`Values`)

In this panel, the method (to obtain different parameter values) and the options relevant to the method are available.

Select method `Vector`.

In the Vector method, one defines the start and end point of the vector (or the start point and the increment), the number of increments, and the corresponding numerical values. The vectors with variable values are calculated equally between the start and end point.

Fill in the number of increments (e.g. 25)



The third panel identifies the part of the code that needs to be executed with different parameter values. One option is to run the complete input file – PHREEQC will be restarted every time. Output files will be overwritten.

A better way is to put the relevant code in a separate file:

Select Specific File

Select `Modules` from the drop down box `Folder`

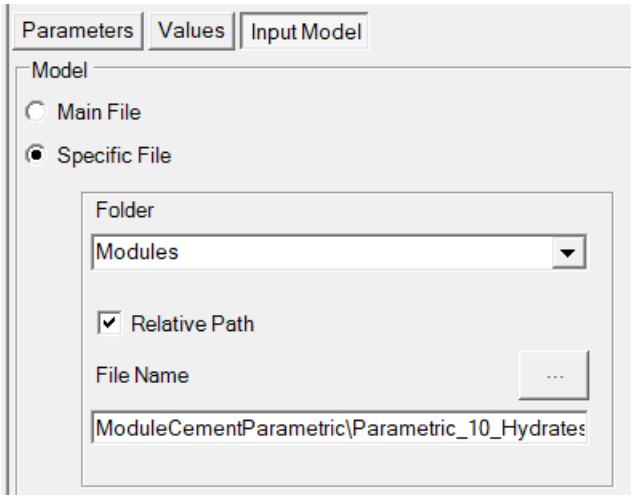
Check `Relative Path`

Click ...

Go to `ModuleCementParametric` Folder

Select `Parametirc_10_Hydrates.hps`

Click `Open`



Step 3 – Set Model and Output Options

Go to the input table via the icon 

Choose model 17-Ca Si Al/fe S Mg C Na K Cl in the `Cement Model` tab – `Elements` group

Go to the `Output` Tab

Details of the output tab is given in **8.1.5** (not available yet).

The first group (`Output – Values`) allows to select the output variables. Hydrates can be reported in moles, volumes (cm^3) and weight (g). Also the saturation index of the hydrates can be send to the output.

The second group (`Output – Figures – General`) allows the select the general settings of the figures. The crucial records are:

`Cement::Figures::Lead$` - define the output variable used for the X-axis. This can be one of the (numerical) variables in the global input table. If _null_, integer values will be used.

Type `GBFS::Amount` in `Cement::Figures::Lead$`

`Cement::Figures::X::Title$` - define the label of the X-axis

Type `GBFS [%]` in `Cement::Figure::X::Title$`

The third group (Output – Figures – Plots) allows to select the type of plots:

`Cement::Figures::Water::Elements::Y::Scale$` - to express concentrations in M, mg/L or both – select both

`Cement::Figures::Solids::Detail$` - not implemented yet – do not change

`Cement::Figures::Solids::Units$` - to express the hydrates for the total system or per 100 g – select both

`Cement::Figures::Solids::Value$` - to define the unit of the hydrates (moles, volume, weight) – select All

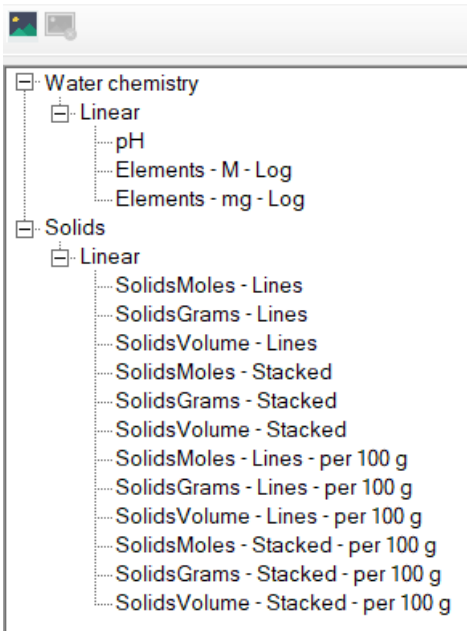
`Cement::Figures::Solids::Type$` - to define the appearance of the plot (lines or stacked lines) – select Both

Run 

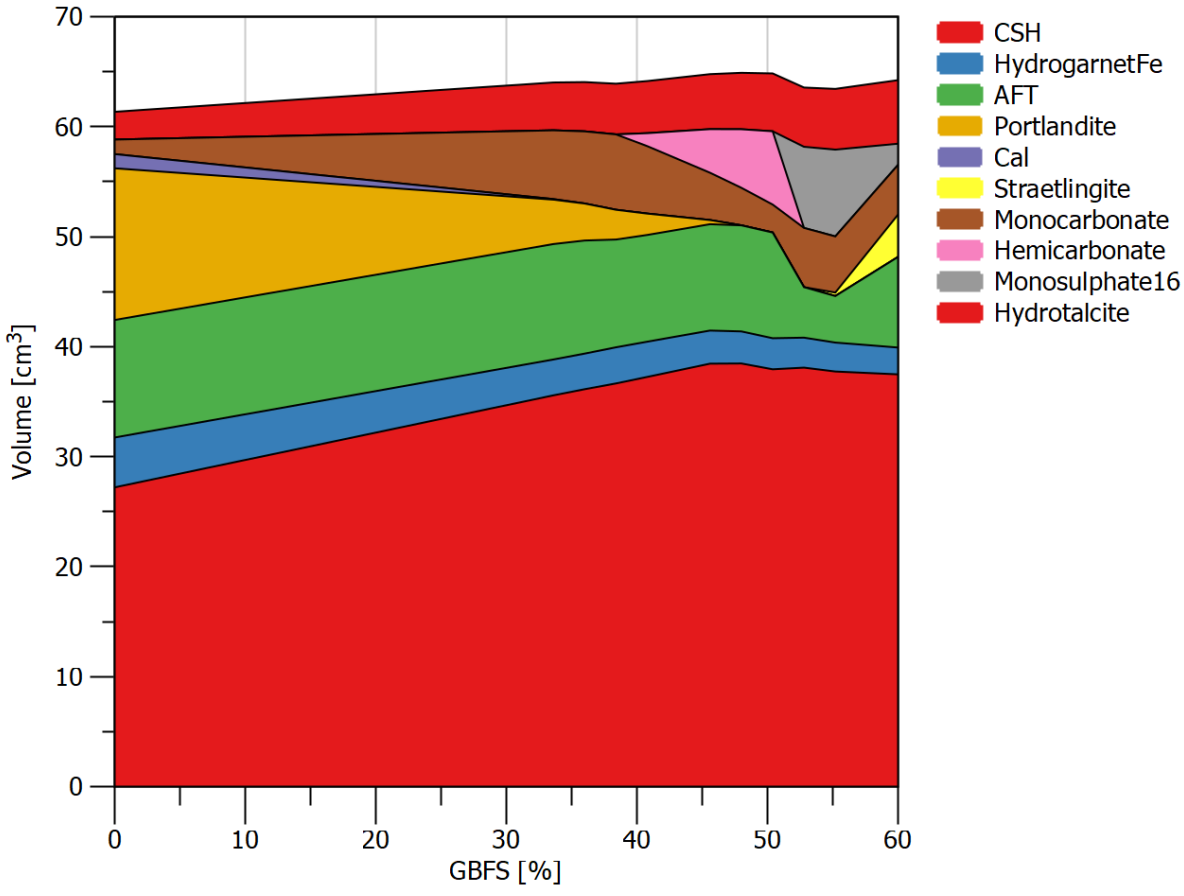
Step 4 – Check Plots

Go to the `Graph Work` Space 

By selecting plots on the list of plots, simulation results can be viewed. The selected options give a range of different plots.



Below is the stacked curve of the volumes of the hydrates as a function of % GBFS. *Need to be done: add initial volume (volume clinkers+volume water), capillary water, gel water, chemical shrinkage. Colors need to be optimized…*



END TUTORIAL 4-5

5. CEMENT HYDRATION

Chapter at a glance

5.1 CEM I – OPC

5.1.1 Reaction extent

5.1.2 Modified Parrot and Killoch – MPK

5.1.2.1 Rate equations

The rate of the hydration of the cement clinkers is described by the set of empirical equations as proposed by Parrot and Killoh (1984) and presented in Lothenbach and Winnefeld (2006). The rate equation for clinker phase *j* is:

$$R_j = \min\left(R_1, R_2, R_3\right) \tag{5.1}$$

With Three equations are used to calculate the rate as a function of hydration degree of the individual clinker (α_t) and the lowest rate is considered as the rate-controlling step:

$$R_1 = \frac{K_1}{N_1}(1-\alpha_t)\left(-\ln(1-\alpha_t)\right)^{(1-N_1)} f_T f_H f_S f_{w/c} \tag{5.2}$$

$$R_2 = \frac{K_2(1-\alpha_t)^{2/3}}{1-(1-\alpha_t)^{1/3}} f_T f_H f_{w/c} \tag{5.3}$$

$$R_3 = K_3(1-\alpha_t)^{N_3} f_T f_H f_{w/c} \tag{5.4}$$

where the first, second and third equations represent, respectively, nuclear and growth, diffusion and formation of the hydration shell. f_T, f_H, $f_{w/c,}$ and f_S are correction factors for, respectively, temperature, hydration degree, water/cement ratio, and Blaine surface area. $f_{w/c}$ is defined as:

$$f_{w/c} = \left(1 + 3.333\left(Hx * \frac{w}{c} - \alpha_t\right)\right)^4 \tag{5.5}$$

if $\alpha_t > H_x * w/c$, otherwise $f_{w/c}$ is 1 with H_x and empirical parameter. The correction factor for temperature, f_T, is defined as:

$$f_T = \exp\left[\frac{E_a}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right] \tag{5.6}$$

with *T* is the temperature, and T_0 is the reference temperature (both in K), and E_A is the enthalpy [J/mol].

Following two functions are not implemented yet:

$$f_H = \left(\frac{RH - 0.55}{0.45}\right)^4 \tag{5.7}$$

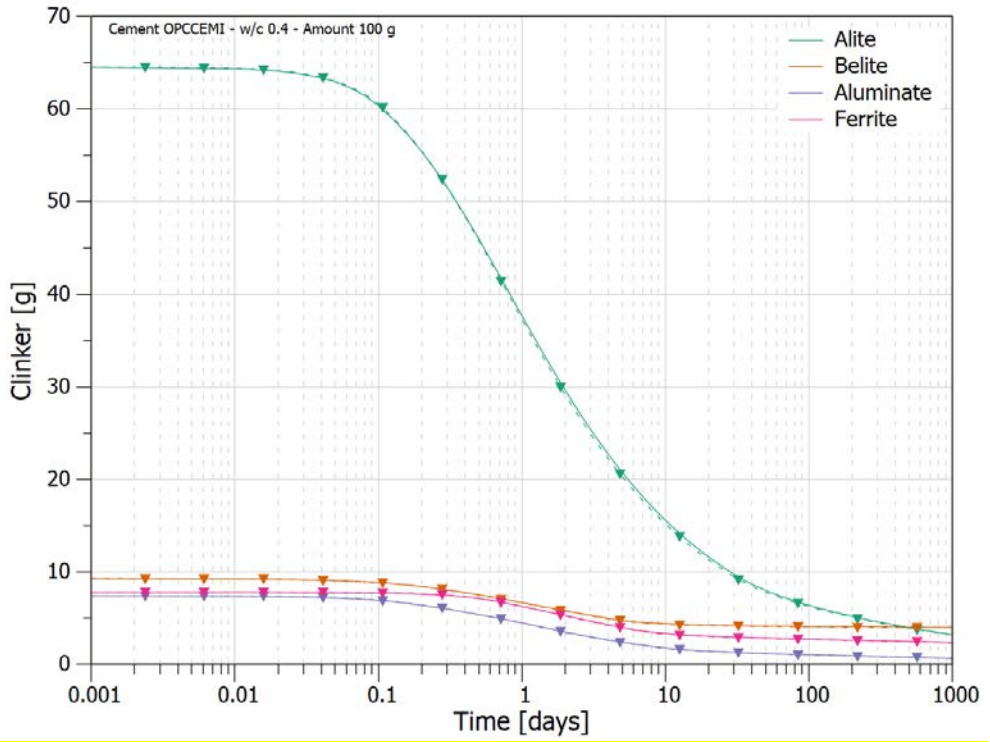
where *RH* is the relative humidity, and:

$$f_s = \frac{A}{A_0} \tag{5.8}$$

with A Blain fineness and A_0 reference area (385 m^2/kg).

The implementation in HPGeochemistry was verified against an excel implementation from B. Lothenbach[1].

[1] Personal communication (D. Kulik, PSI, November 2020)



need to recheck

5.1.2.2 Implementation

The hydration of the OPC clinker is done in three steps. Hereafter, the steps are discussed for the hydration of 100 cement of CEM I (100% OPC clinkers) with a w/c of 0.5 and a temperature of 100 °C.

Definition of OPC in terms of constituents

Template: `Hydration_10_OPC_minerals`

When executed, the module write the input file based on the information in the `Global Definitions` input workspace. The input file is saved as `hydration._in` in the simulation directory.

STEP 1 – FIGURE 5-1

- Line 1 includes the scripts for the modified Parrot and Killoch rate equations (Eqs. (5.2) to (5.8)) from the cement module.
- Line 3-5 defines the aqueous phase for the hydration – the amount of water is calculated from the water cement ratio (`Cement::Recipe::w_c`) and the amount of cement (`Cement::Recipe::Amount`) in the `Cement Recipe – mineral` table. The temperature is obtained from `Cement::Recipe::T`.

```
1  INCLUDE$ ModuleCement_PhreeqcInput_20_Hydration_10_OPC_01_mPKrates.hps
2  END
3  SOLUTION 1
4  -water 0.05
5  -temperature 25
6  PRINT
7  -reset   false
8  SELECTED_OUTPUT
9  -active false
10 END
```

Figure 5-1 – Hydration of OPC clinkers using modified Parrot-Killoch equations: Extract of hydration._in – Definition of the solution

STEP 2 – FIGURE 5-2

- Line 12 includes the file of the cement model using the information of the `Cement Model` table.
- Line 13-19 defines the amounts of the constituents in the OPC clinkers that react immediately. The **`REACTION`** entity uses following input:
  - The composition of OPC in terms of constituents is defined in the data block `OPC_min`.
  - For each of the constituents that are considered to react immediately (all constituents in `OPC_min` except the four clinker phases alite, belite, aluminate and ferrite), the formula and reaction amounts are added. The reaction amount is obtained by .
  - Only constituents with an amount larger than 0 are added to **`REACTION`**.
  - The total reaction amount is 1 (line 19) as total added reaction components were defined.
- Line 20-22 saves the resulting solution and phase compositions for step 3.

```
11 USE SOLUTION 1
12 INCLUDE$ CementModel1._in
13 REACTION
14 CaO 0.00820295
15 CaSO4 0.03085
16 CaCO3 0.0439618
17 Na2SO4 0.000675846
18 K2SO4 0.00918162
19 1
20 save solution 1
21 save equilibrium_phases 1
22 save solid_solution 1
23 END
```

Figure 5-2 – Hydration of OPC clinkers using modified Parrot-Killoch equations: Extract of hydration._in – Equilibration with soluble components

STEP 3 – FIGURE 5-3

- Line 24-26 indicate to use the save solution and phase composition from step 2.
- Line 27-45 defines the **`KINETICS`** entity. The **`KINETICS`** entity uses following input:
  - Formule (e.g. line 29 `-formula`) – stoichiometric coefficients of the different oxides calculated based on the weigh percentage defined in datablocks `OPC_alite`, `OPC_belite`, `OPC_aluminate` and `OPC_ferrite`. The record in the data block is obtained by the selected record in the drop down menus in the `Cement Recipe – mineral` table (records `Cement::Clinker::Alite$`, `Cement::Clinker::Belite$`, `Cement::Clinker::Aluminate$`, `Cement::Clinker::Ferrite$`). The weight of oxide in 1 gram of clinker is multiplied with the molar mass of the clinker (column `MolarMass` in the datablock) and divided by the gram formula weight.

- Amount (e.g. line 30 `-m0`) – the amount is calculated from the information in the `Cement Recipe – mineral table`. The mass of cement (`Cement::Recipe::Amount`) is multiplied by the clinker amount (`Cement::Clinker::Amount`, in weight percent – value divided by 100) and divided by the molar mass of clinker (column with heading `MolarMass` in the data blocks `OPC_alite`, `OPC_belite`, `OPC_aluminate` and `OPC_ferrite`). Note that the amount of reacted clinker (`Cement::Clinker::ReactedAmount`) is not used for the hydrations calculations.
- Parameters (e.g. line 31 `-parms`) – lists the modified Parrot and Killoch parameters and the water/cement ratio. The list of parameters is thus: K_1, N_1, K_2, K_3, N_3, Hx, E_a, E_{aT}, w/c. The first eight parameters are from the tables `OPC_alite_PK`, `OPC_belite_PK`, `OPC_aluminate_PK` and `OPC_ferrite_PK`. The record is selected via the drop down menus in the `Hydration` table (`Cement::Hydration::Alite::Par$`, `Cement::Hydration::Belite::Par$`, `Cement::Hydration::Aluminate::Par$`, `Cement::Hydration::Ferrite::Par$`). The w/c is defined in the `Cement Recipe – mineral` table (`Cement::Recipe::w_c`).
- Line 44 `-step` : lists the incremental time steps over which the rate equations are integrated. By default, time steps for the first day are defined between 10^{-3} day and 1 day at time steps 10^a with values of a between -3 and 0 with a step of 0.1. Next, similarly time steps 10^a are taken between 0 and the log of the final hydration time defined in the `Hydration` table (`Cement::Hydration::Time`) where the step interval is calculated based on the number of steps (`Cement::Hydration::Steps` in the `Hydration` table).

```
24 use solution 1
25 use equilibrium_phases 1
26 use solid_solutions 1
27 Kinetics
28 OPCAlite
29 -formula H 0  CaO 2.91112 SiO2 0.956256 Al2O3 0.0223614 SO3 0 MgO 0.0622265 Fe2O3 0.00999448 Na2O 0.00367867 K2O 0.00242048
30 -m0 0.241228
31 -parms 1.5 0.7 0.05 1.1 3.3 2 41570 20  0.5
32 OPCBelite
33 -formula H 0  CaO 1.94767 SiO2 0.901733 Al2O3 0.0354252 SO3 0.00214825 MgO 0.0213376 Fe2O3 0.00969389 Na2O 0.00277513 K2O 0.0164338
34 -m0 0.0872093
35 -parms 0.5 1 0.02 0.7 5 1.55 20785 20  0.5
36 OPCAluminate
37 -formula H 0  CaO 2.72476 SiO2 0.166242 Al2O3 0.828721 SO3 0 MgO 0.0937724 Fe2O3 0.0862178 Na2O 0.0435567 K2O 0.0200616
38 -m0 0.0292636
39 -parms 1 0.85 0.04 1 3.2 1.8 54040 20  0.5
40 OPCFerrite
41 -formula H 0  CaO 5.46513 SiO2 0.386577 Al2O3 1.38581 SO3 0 MgO 0.480245 Fe2O3 0.86464 Na2O 0.01041 K2O 0.0136991
42 -m0 0.0125542
43 -parms 0.37 0.7 0.015 0.4 3.7 1.65 34087 20  0.5
44 -step 1E-3 0.001 0.000258925 0.000325968 0.000410369 0.000516624 0.000650391 0.000818794 0.0010308 0.0012977 0.00163371 0.00205672 0.00258925
45 incremental_reactions true
```

Figure 5-3 – Hydration of OPC clinkers using modified Parrot-Killoch equations: Extract of hydration._in – Kinetic reactions of the four clinkers

OUTPUT

Tabulated data: HydrationResults.out …

Graphical data: ….

5.1.2.3 Hydration of CEM I – OPC

TUTORIAL 5-1 HYDRATION OF CEM I – OPC USING MODIFIED PARROT AND KILLOCH EQUATIONS

Description

Simulate the hydration of CEM-I (100% OPC clinkers) of 100 g OPC clinkers with a w/c ratio of 0.5 at 25°C for 1000 days.

You will learn to

Use the template to simulate hydration of OPC CEM-I with the modified Parrot-Killoch parameters

Depends on

Tutorial 3-2 Create managed project with the cement module

Step 1 – Create the simulation from the template to calculate hydration of OPC clinkers

Go to `Project Manager` 

Select the project in the bottom panel

Go to `Templates` tab in the top panel

Select the template "`Hydration_10_CEM-I`"

Click the "`Create new from template`" button 

The `Metadata` dialogue window opens.

Define the name of the simulation (`Hydration CEM-I`)

Click `OK`

The Project Manager is active again.

Select the simulation (`Hydration CEM-I`) and click `Select`

Step 2 – Define the hydration process

Go to `Global Definitions`

Go to `Global Variables`

Go to `Cement Model` tab

Select `system 17` (Ca-Si-Al-Fe-S-Mg-C-Na-K-Cl) from record `Cement::Model::System$`

Select `s_s_ideal` for `Cement::Model::AFT_s_s$`

Cement recipe uses the default values

Go to `Hydration` tab

Change `Cement::Hydration::Time` to `1000`

Run ()

Results

Go to the `Figure` workspace ()

Select `Solids – Logscale – SolidsVolume-Stacked`

Go to `Edit Graph` tab

Type:

```
wc = sprintf("%g",Cement_Recipe_w_c)
txt = "CEM-I\nw/c ".wc
set title txt
```

These commands are gnuplot statements. All gnuplot statements that are not part of the plot statement can be used to customize the appearance of the plot.

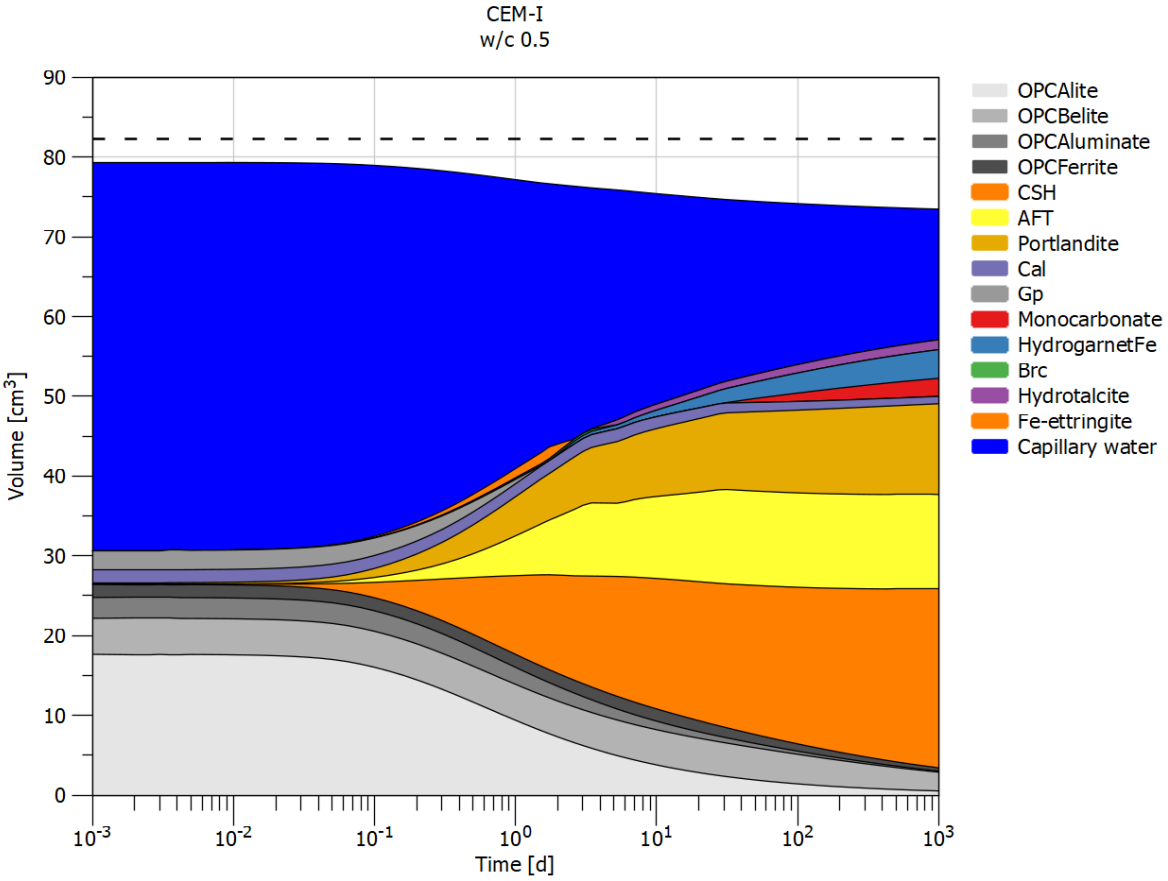


Figure 5-4 - Hydration of OPC clinkers using modified Parrot-Killoch equations:

Need to check – is water in gelpores of CSH in volume of CSH

The horizontal dashed line is the initial volume calculated as the mass of OPC clinker (g) divided by its bulk density (in the table in kg/dm³ or g/cm³). The difference between this line and the sum of the volumes of the clinkers, capillary water and hydrates (i.e., the space between the line and the capillary water) is the chemical shrinkage. The chemical shrinkage at time 10^{-3} d is due to the dissolution of the constituents that dissolved immediately (step 2 of the implementation in paragraph 0).

END TUTORIAL 5-1

5.1.3 Shrinking-core unit-cell approach

5.2 Supplementary cementitious materials

5.2.1 Based on mPK rate equations

5.2.1.1 Rate equations

It is assumed that the processes determining the rate of hydration of SCM are the same as for the clinkers of OPC which is reported in several studies (REF). The same set of equations as presented in section is used, with following two adaptations to the general rate Eq. (5.1): (i) the fraction of easily soluble

(amorphous, poorly crystalline fraction) phases (R_F with a value between 0 and 1) which will hydrate according to the adapted rate equations; the remaining fraction (crystalline fraction) is inert[2], and (ii) a factor (f_{CH}) to express the time-dependent availability of the precursor for SCM being portlandite for the pozzolanic SCMs. R_F is potentially different from 1 for FA (e.g., …) but assumed to be 1 for e.g. SF (e.g., …). The second factor… The rate equation for SCMs phases is thus:

$$R_j = R_F \min(R_1, R_2, R_3) f_{CH} \quad (5.9)$$

where *R*1, *R*2, and *R*3 are given by, respectively, Eq. (5.2), (5.3), and (5.4).

No - Absolute – fractional

Linear - exponential

5.2.1.2 Hydration of OPC blended with fly-ash

Glosser et al. (2020) proposed to use the mPK rate equations to model the hydration of fly-ash in blended cement systems. The mPK rate equations are defined for three oxide constituents of fly-ash (SiO_2, Al_2O_3 and CaO) and parameters are given in their table 4. Note that there is no information on temperature dependence. Their comparison with a mPK implementation with a uniformly dissolving fly-ash composition (thus congruently reacting fly-ash).

Ratio of amorpheous (reactive) versus crystalline (non reactive) oxides – not implemented

Not clear what to do with other oxides in FA -> not released yet

Check also https://www.sciencedirect.com/science/article/pii/S0958946520303541?via%3Dihub

Implementation

Template: `Hydration_20_SCM_01_FA_OPC_minerals`

Hydration_20_SCM_01_FA_OPC_minerals

Tutorial

TUTORIAL 5-2 HYDRATION OF OPC BLENDED WITH FLY-ASH USING MODIFIED PARROT AND KILLOCH EQUATIONS

Description

Simulate the hydration of a blended cement with 80% CEM-I and 20% fly-ash for 100g cement with a w/c ratio of 0.5 at 25°C for 1000 days.

You will learn to

Depends on

Step 1 – Create the simulation from the template to calculate hydration of OPC clinkers

Go to `Project Manager`

Select the project in the bottom panel

Go to `Templates` tab in the top panel

[2] Or optionally very slowly dissolving – to be discussed. This fraction could be assumed to dissolve e.g. following TST theory with very slow rate which could be assumed e.g. for aggregates as well, to be discussed.

Select the template "Hydration_20_SCM_01_FA_OPC_minerals"

Click the "Create new from template" button 

The Metadata dialogue window opens.

Define the name of the simulation (Hydration OPC-FA)

Click OK

The Project Manager is active again.

Select the simulation (Hydration OPC-FA) and click Select

Step 2 – Define the hydration process

Go to Global Definitions

Go to Global Variables

Go to Cement Model tab

Select system 13 (Ca-Si-Al-S-Mg-C-Na-K-Cl) from record Cement::Model::System$

Select s_s_ideal for Cement::Model::AFT_s_s$

Go to Cement Recipe – mineral tab

Change amount of clinker (Cement::Clinker::Amount) to 80 and fly-ash (FA::Amount) to 20

Go to Hydration tab

Change Cement::Hydration::Time to 1000

Step 3 – Make title for each graph

Go to GeoChemistry tab

Add following code before first line

```
SELECTED_GRAPH
-start
wc = sprintf("%g",Cement_Recipe_w_c)
opc = sprintf("%g",Cement_Clinker_Amount)
fa = sprintf("%g",FA_Amount)
txt = "Blended cement: OPC ".opc."% - FA ".fa."%\nw/c ".wc
set title txt
-end
```

Add explanation

Run ()

Results

Go to the Figure workspace ()

Select Solids – Logscale – SolidsMoles-Stacked

Gives warning -> To Check

Only figure with moles are correct – others need to be implemented

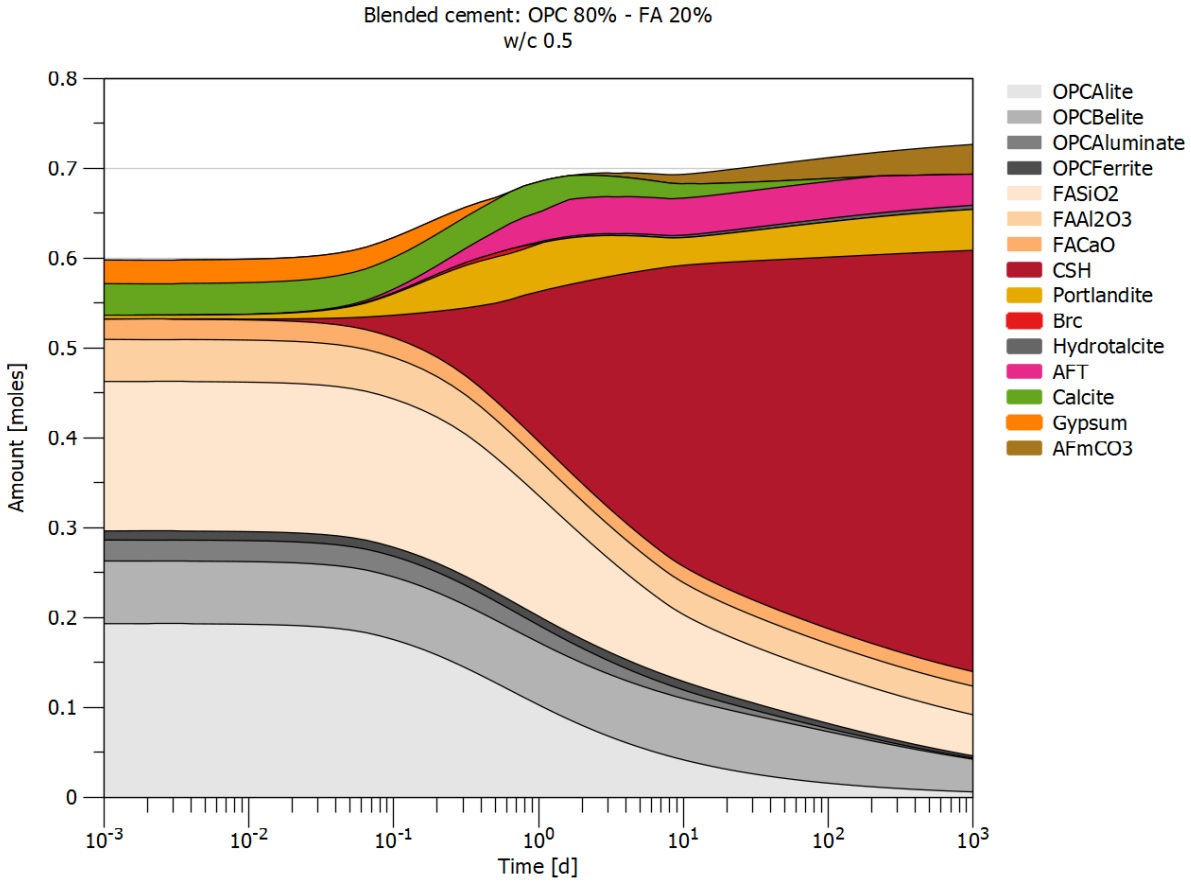


Figure 5-5 - Hydration of Blended OPC clinkers (80%) and fly ash (20%) using modified Parrot-Killoch equations:

END Tutorial 5-2

6. MODELLING DEGRADATION PROCESSES – THERMODYNAMIC APPROACHES (MIXING TANK APPROACHES)

Chapter at a glance

6.1 Ingress

TUTORIAL 6-1 CARBONATION OF CEM-I

Description

Simulate the carbonation of CEM-I (100% OPC) of 100 g CEM-I with a water/cement ratio of 0.5 and at a temperature of 25°C. The CEM I oxide composition is taken from the Lothenbach and Winnefeld (2006) (their table 1). Calculations are done in the Ca-Si-Al-S-Mg-C system.

You will learn to

Implement the ingress process with the ingress template

Depends on

Tutorial 3-2 Create managed project with the cement module

Step 1 – Create the simulation from the template to blend materials

Go to `Project Manager`

Select the project in the bottom panel

Go to `Templates` tab in the top panel

Select the template "`Hydrates_20_Ingress_general`"

Click the "`Create new from template`" button

The `Metadata` dialogue window opens.

Define the name of the simulation (`Carbonation`)

Click `OK`

The Project Manager is active again.

Select the simulation (`Carbonation`) and click `Select`

Step 2 – Define the carbonation process

Go to `Global Definitions`

Go to `Cement Model` tab

`Cement::Model::System$` change to system 13

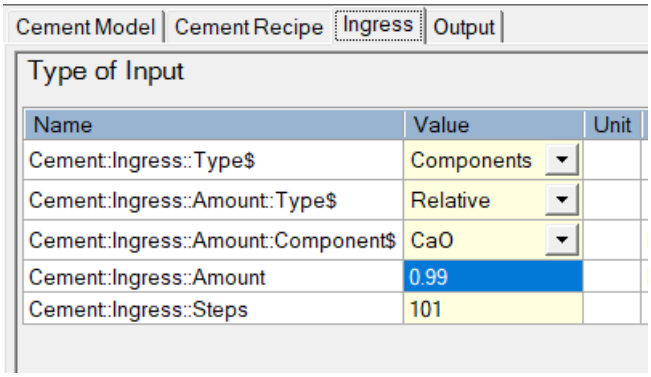
Change the AFt non-ideal model to ideal model (otherwise, PHREEQC does not convergence when AFt is almost completely dissolved):

`Cement::Model::AFT_s_s$` change to `s_s_ideal`

Go to `Ingress` tab

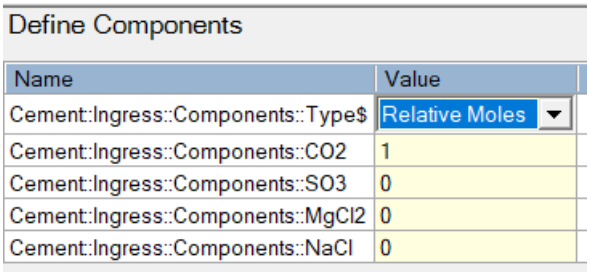
In the first group (Type of Input), the type of ingress can be defined (only Components is implemented). With the selection Components, the components in the group Define Component will be added to the system. The total amount of added components is defined in absolute terms (not implemented) or relative terms to one of the oxides in the cement system. In this particular case, CO2 will be added to 0.99% of the amount of CaO (more will lead to non-convergence) in 101 steps.

`Cement::Ingress::Amount` 0.99



In the second group (Define Components), the relative amount of a component is defined – here the relative amount of CO_2 per mol of CaO is defined.

`Cement::Ingress::Components::CO2` 1



Step 3 – Plots

Go to the `Output` tab and select the desired plots

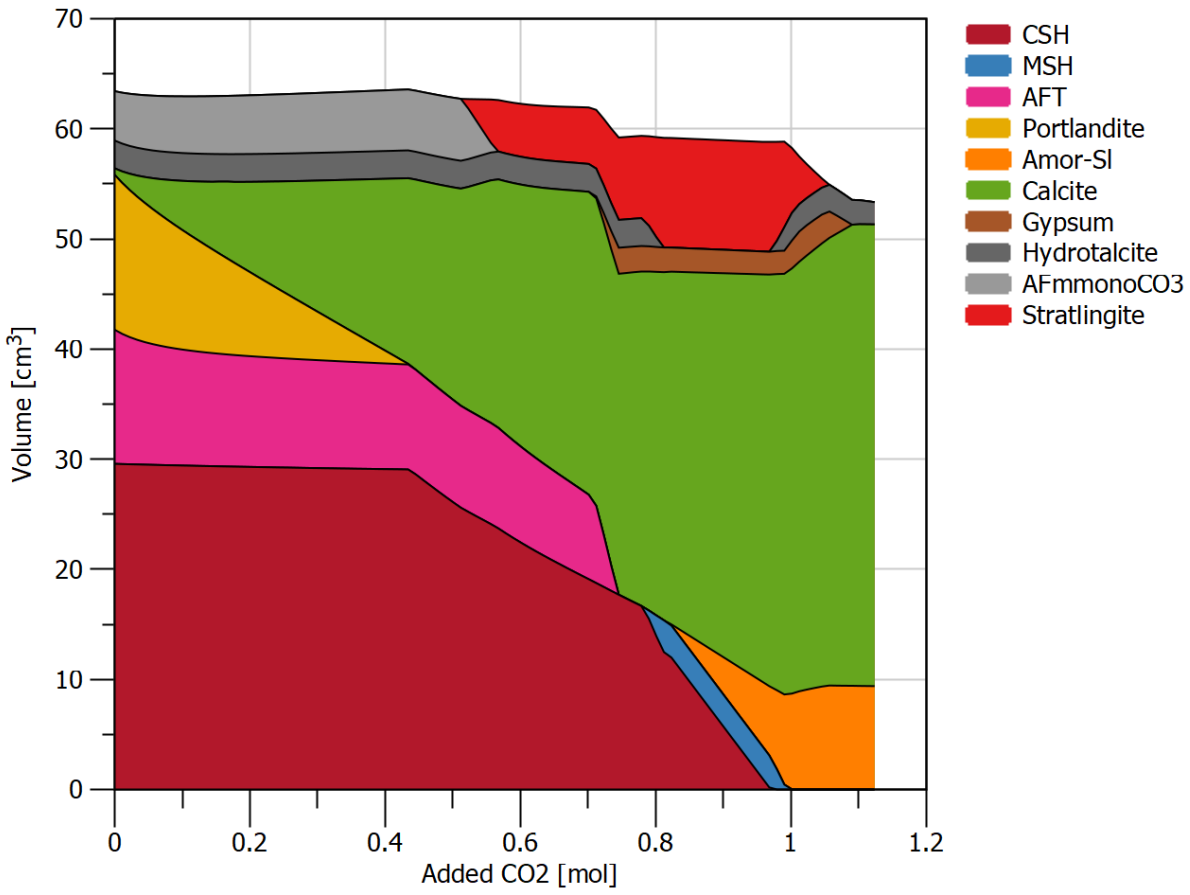
Change `Cement::Figures::Lead$` to `Cement::Ingress::Components::CO2` to have the lead variable (X-axes) the absolute amount of CO_2 that have been added.

Change `Cement::Figures::X::Title$` to `Added CO2 [mol]`

Run

Output

Example figure: Stacked plot of solids expressed in volumes



END TUTORIAL 6-1

6.2 Leaching

TUTORIAL 6-2 LEACHING OF CEM-I WITH A SOLUTION AT PH 3

Description

Simulate the leaching of CEM-I (100% OPC) of 100 g CEM-I with a water/cement ratio of 0.5 and at a temperature of 25°C. The leaching solution has a pH of 3 (bufferd by HCl) – assume a cumulative leaching to 500 L of water. The CEM I oxide composition is taken from the Lothenbach and Winnefeld (2006) (their table 1). Calculations are done in the Ca-Si-Al-S-Mg-C-Na-K system.

You will learn to

Depends on

Tutorial 3-2 Create managed project with the cement module

Step 1 – Create the simulation from the template to blend materials

Go to `Project Manager`

Select the project in the bottom panel

Go to `Templates` tab in the top panel

Select the template "Hydrates_30_Leaching_general"

Click the "Create new from template" button 

The Metadata dialogue window opens.

Define the name of the simulation (Leaching)

Click OK

The Project Manager is active again.

Select the simulation (Leaching) and click Select

Step 2 – Define the leaching process

Go to Global Definitions

Go to Global Variables

Go to Cement Model tab

Cement::Model::System$ change to system 13

Change the AFt non-ideal model to ideal model (otherwise, PHREEQC does not convergence when AFt is almost completely dissolved):

Cement::Model::AFT_s_s$ change to s_s_ideal

Go to Leaching tab

In the first group (Leaching amounts), the cumulative amount of leaching is defined in four steps of respectively increments of 0.01, 0.1, 1 and 10 L. The default values leach first 1 L with 0.01 L increments, 9 L with 0.1 L and 90 L with 1 L, in total 100 L thus.

Adapt the fourth step (10 L) to have a total amount of 400 L

Cement::Leaching::Step 390

The cement::Leaching::Type$ is selected on Components. The composition of the leaching solution is defined in terms of concentration of different components given in the second group Components. Default values are 1E-15M – this will avoid to have warnings during calculations linked to the presence of phases in the model without the corresponding elements in solution.

Change Cement::Leaching::Cl to 1E-3

Change Cement::Leaching::pH to 3

Select Cl in the drop down list Cement::Leaching::Charge$

The solution thus have a pH of 3 and Cl concentration is adapted to have a charge-balanced solution. This will result in following PHREEQC input during calculations (for increment of 0.01L):

```
solution 169898
-water 0.01
-units mol/kgw
Ca 1e-15
Al 1e-15
pH 3
Cl 0.001 charge
C 1e-15
K 1e-15
S 1e-15
Na 1e-15
Mg 1e-15
Si 1e-15
```

Step 3 – Plots

Go to the `Output` tab and select the desired plots

Group `Output – Figures - General`

Change `Cement::Figures::X::Title$` to `Added water [kg]`

Change `Cement::Figures::X::Scale$` to `Logscale` or `Both`

Change `Cement::Figures::Y::Grid$` to `Both Log`

Group `Output – Figures – Plots`

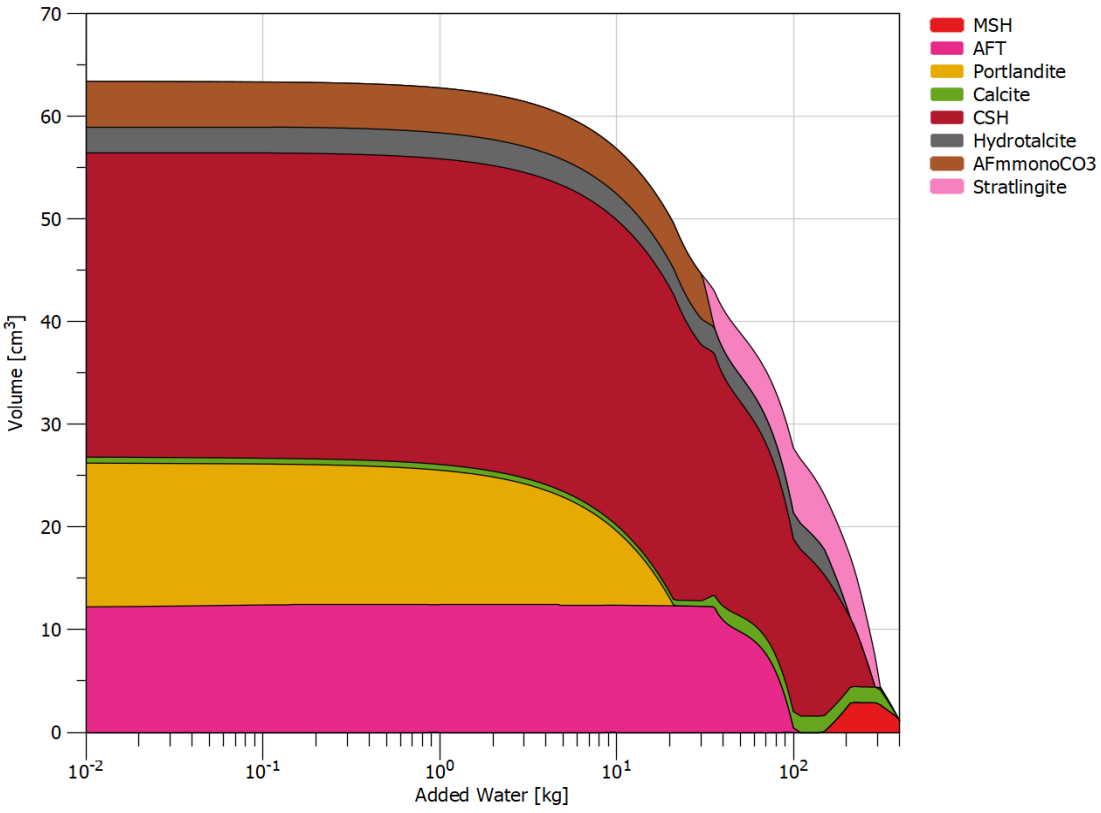
Change `Cement::Figures::Solids::Value$` to `All`

Change `Cement::Figures::Solids ::Types$` to `Both`

Run

Output

Example figure: Stacked plot of solids expressed in volumes



END TUTORIAL 6-2

7. COUPLED REACTIVE TRANSPORT MODELLING – CONTINUUM SCALE

Chapter at a glance

7.1 One-dimensional coupled reactive transport models – diffusive models

Templates are provided for following conditions

- Constant boundary condition without continuing hydration reactions
- Constant boundary condition with continuing hydration reactions
- Submerged boundary condition with

7.1.1 Constant boundary condition without continuing hydration reactions

1D diffusive transport with:

- Closed boundary condition at one side
- Constant boundary condition at other side

8. CEMENT MODULE

8.1 Global Variables tables

8.1.1 Cement Model

8.1.2 Cement Recipes

8.1.2.1 Cement Recipe – mineral

8.1.3 Hydration

8.1.3.1 Hydration

8.1.4 Processes

8.1.5 Output (blending, ingress, leaching)

8.2 Data Blocks

8.2.1 Composition of OPC

8.2.1.1 In terms of oxides: OPC

8.2.1.2 In terms of constituents: OPC_min

The data block `OPC_min` defines OPC clinkers in terms of its constituents.

The first six columns (column index 0 to 5) cannot be changed in terms of variable name or position in the data block. The definition of the columns are:

`Name`	Unique name of the specific composition. The name is a selection option in the `Cement::Clinker::Mineral::Reference$` record in the Cement Recipe – mineral table.
`Unit`	Defines the used unit. Only mass% (i.e. g/100 g OPC clinker) is valid. Not used.
`Alite`	The mass% of alite in the OPC cement clinker.
`Belite`	The mass% of belite in the OPC cement clinker.
`Aluminate`	The mass% of aluminate in the OPC cement clinker.
`Ferrite`	The mass% of ferrite in the OPC cement clinker.

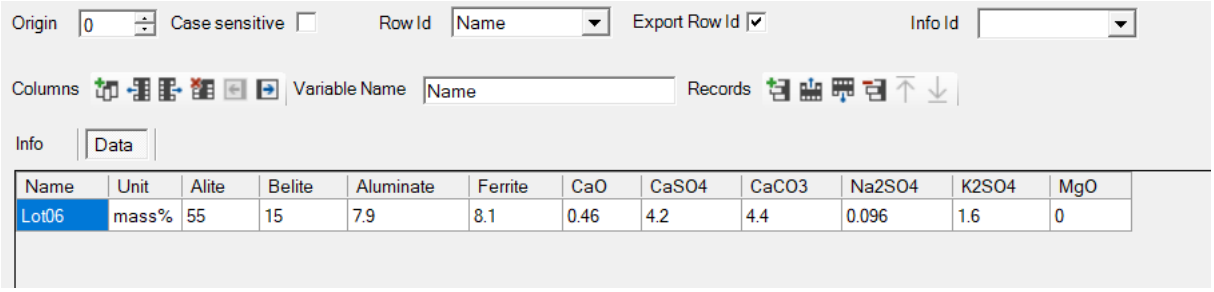


Figure 8-1 – OPC_min Data Block – Composition of OPC clinkers in terms of constituents

8.2.1.3 OPC_alite, OPC_belite, OPC_aluminate, OPC_ferrite

8.2.2 Composition of SCM

8.2.2.1 In terms of oxides

8.2.3 Hydration kinetics

8.2.3.1 Parameters for modified Parrot and Killoch parameters – OPC_alite_PK, OPC_belite_PK, OPC_aluminate_PK, OPC_ferrite_PK

9. ADVANCED METHODS

9.1.1 Set-up a template simulation

To ease implementation of the different simulations discussed in this chapter, a template simulation is made. Other simulations will be based on this template.

The template will contain (i) the input tables for the selection of the cement model and the definition of the cement recipe; (ii) data blocks with the oxide composition of OPC and SCMs; and (iii) the required input for the equilibrium hydration model.

TUTORIAL 9-1 CREATE TEMPLATE FOR CEMENT MODEL AND RECIPE DEFINITION

Problem Definition

Here we blab la bla

Define a new

TUTORIAL 9-2 CREATE TEMPLATE FOR CEMENT MODEL AND RECIPE DEFINITION

Problem Definition

Here we blab la bla

Define a new

TUTORIAL 9-3 WHY A SECOND PROBLEM

Problem Definition

REFERENCES

Glosser, D., Suraneni, P., Isgor, O.B. and Weiss, W.J. 2020. Estimating reaction kinetics of cementitious pastes containing fly ash. Cement and Concrete Composites 112, 103655.

Glosser, D., Suraneni, P., Isgor, O.B. and Weiss, W.J. 2021. Using glass content to determine the reactivity of fly ash for thermodynamic calculations. Cement and Concrete Composites 115, 103849.

Lothenbach, B. and Winnefeld, F. 2006. Thermodynamic modelling of the hydration of Portland cement. Cement and Concrete Research 36, 209-226.

Parrot, L.J. and Killoh, D.C. 1984. Prediction of cement hydration. British Ceramic Proceedings 35, 41-53.

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